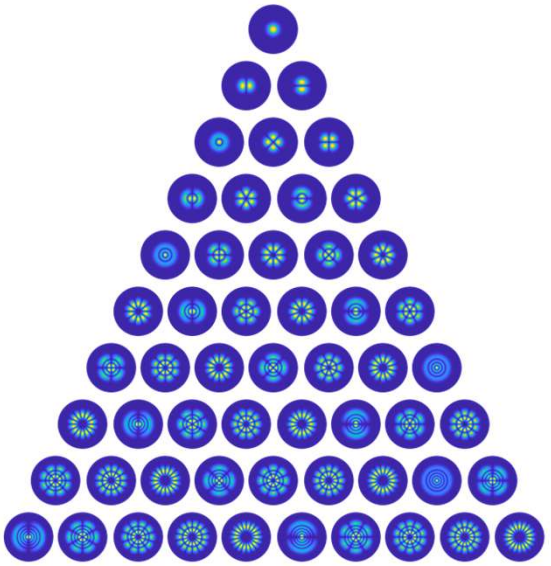
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

Lecture 8

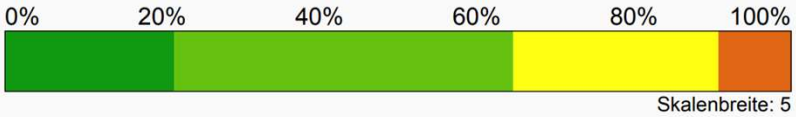
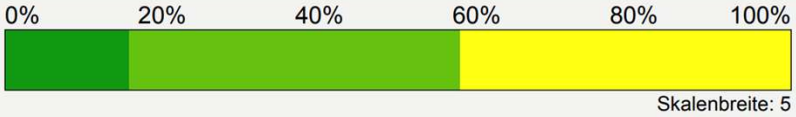
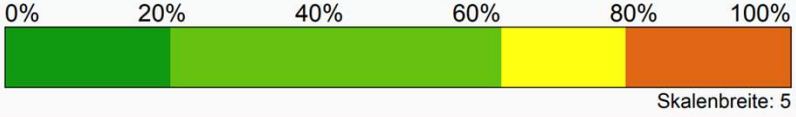
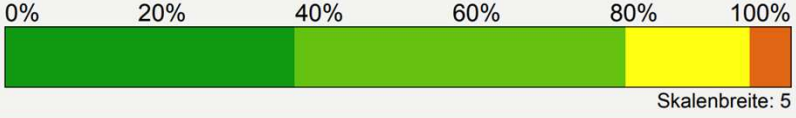


Prof. Dr.-Ing. Georg Rademacher

Organisational matter

University of Stuttgart
Germany

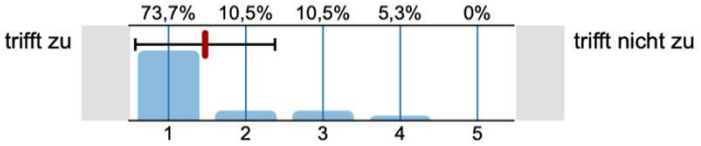
- Next week, we will swap lecture and excersise
- -> Lecture on Tuesday from 2 – 3.30
- -> Excersise on Thursday from 2 – 3.30

	+ -	Ø
INDEX: Zufriedenheit mit der Lehrveranstaltung (Skalenbreite: 5)		2,23
Index Item 1: Die Lehrveranstaltung ist außerordentlich gut organisiert.		2,26
Index Item 2: Der inhaltliche Aufbau der Lehrveranstaltung ist für mich nachvollziehbar.		2,42
Index Item 3: Die Lernziele der Lehrveranstaltung werden klar kommuniziert.		2,26
Index Item 4: Die Inhalte werden verständlich vermittelt.		2,37
Index Item 5: Ich werde zum Mitdenken motiviert.		1,89

	+	-	Ø
Index Item 6: Es wird auf die Anliegen und Belange der Studierenden eingegangen.	0% 20% 40% 60% 80% 100% Skalenbreite: 5		2,18
Index Item 7: Ich habe durch die Teilnahme an dieser Lehrveranstaltung viel gelernt.	0% 20% 40% 60% 80% 100% Skalenbreite: 5		2,22

Stuttgart

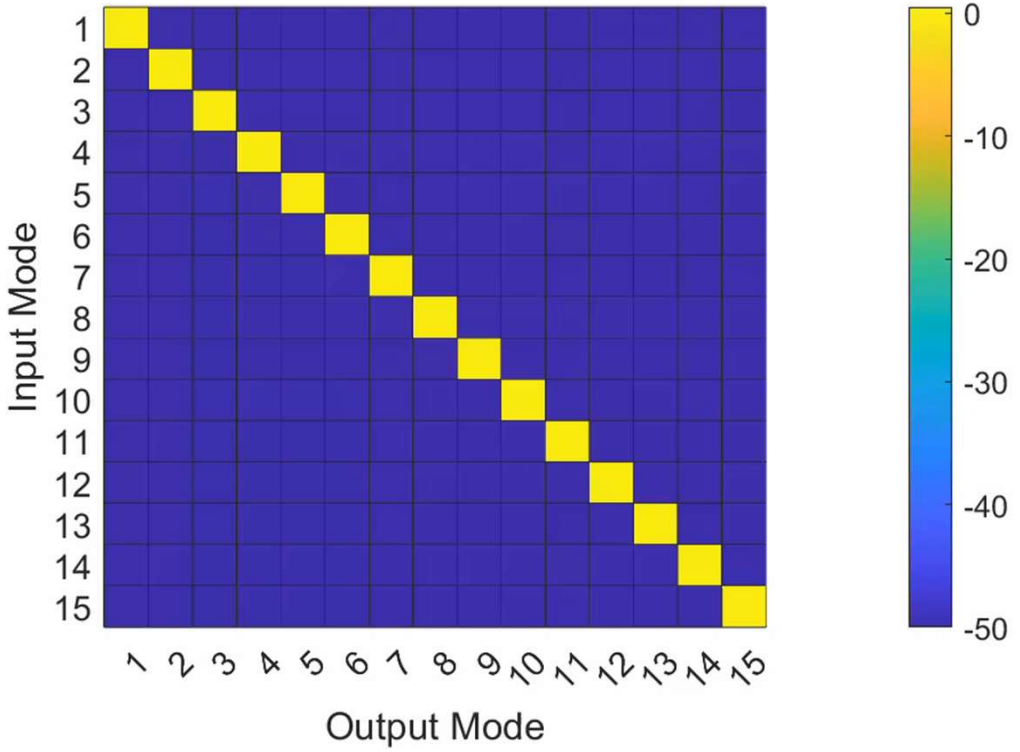
6.1) How much do you like the bonus tests?



- It's better to give practical live examples rather than only theory lectures. I enrolled by thinking practical experience to gain but it's completely theory.
- Like to have some assignments and give more problems related to each exercise.

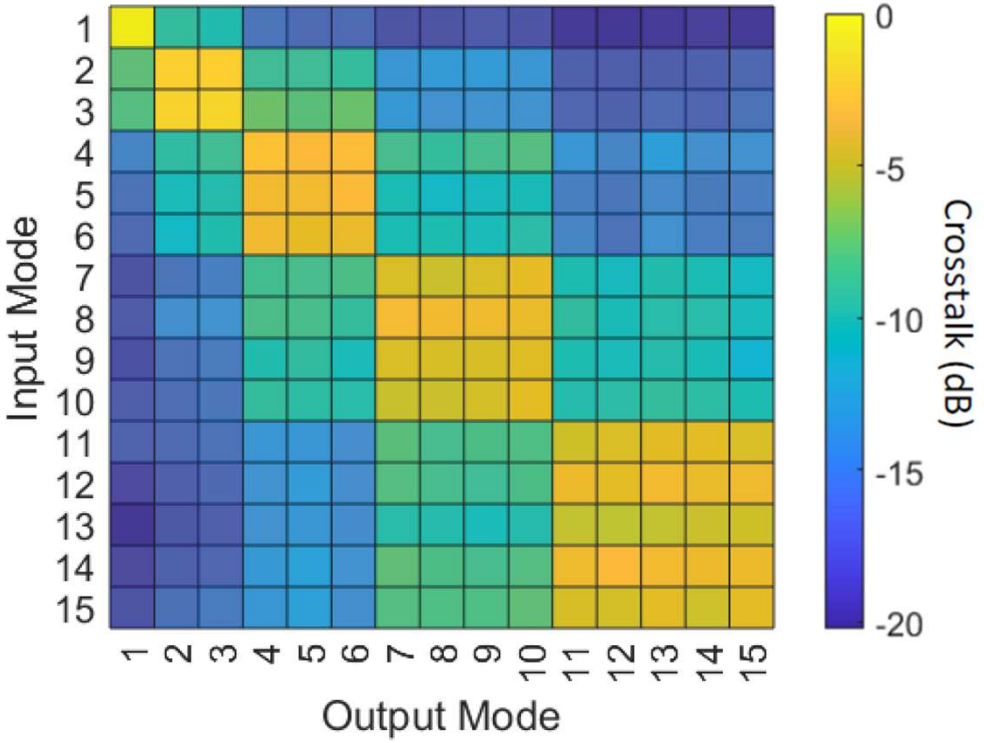
Simulation: Crosstalk vs. offset

University of Stuttgart
Germany

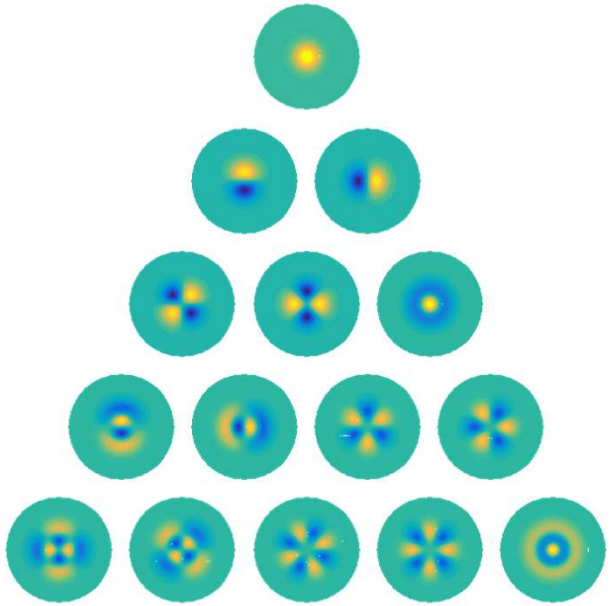




Experiment: Crosstalk



University of Stuttgart
Germany



Butt-coupling of (SI) SMF

$0, \mu \neq \nu$

University of Stuttgart
Germany

$$\iint_A \psi_e \cdot \psi_\nu^* \, dA = \sum_\mu c_\mu \iint_A \psi_\mu \cdot \psi_\nu^* \, dA \quad \Rightarrow \quad c_\mu = \frac{\iint_A \psi_e \, \psi_\mu^* \, dA}{\iint_A \psi_\mu \cdot \psi_\mu^* \, dA}$$

$$\mu = 0 \rightarrow c_\mu = c_0$$

$$P_e = \frac{1}{2} \iint_A E_x \cdot H_y^* \, dA \approx \frac{1}{2 Z_F} \iint_A |\psi_e|^2 \, dA, \qquad P_0 = \frac{1}{2 Z_F} \iint_A |c_0 \cdot \psi_0|^2 \, dA$$

$$\eta = \frac{P_0}{P_e} = |c_0|^2 \frac{\iint_A |\psi_0|^2 \, dA}{\iint_A |\psi_e|^2 \, dA} = \frac{\left| \iint_A \psi_e \psi_0^* \right|^2}{\iint_A |\psi_e|^2 \, dA \cdot \iint_A |\psi_0|^2 \, dA}$$

Butt-coupling of (SI) SMF

University of Stuttgart
Germany

$$\psi_e = \exp\left(-\frac{r^2}{w_1^2}\right)$$

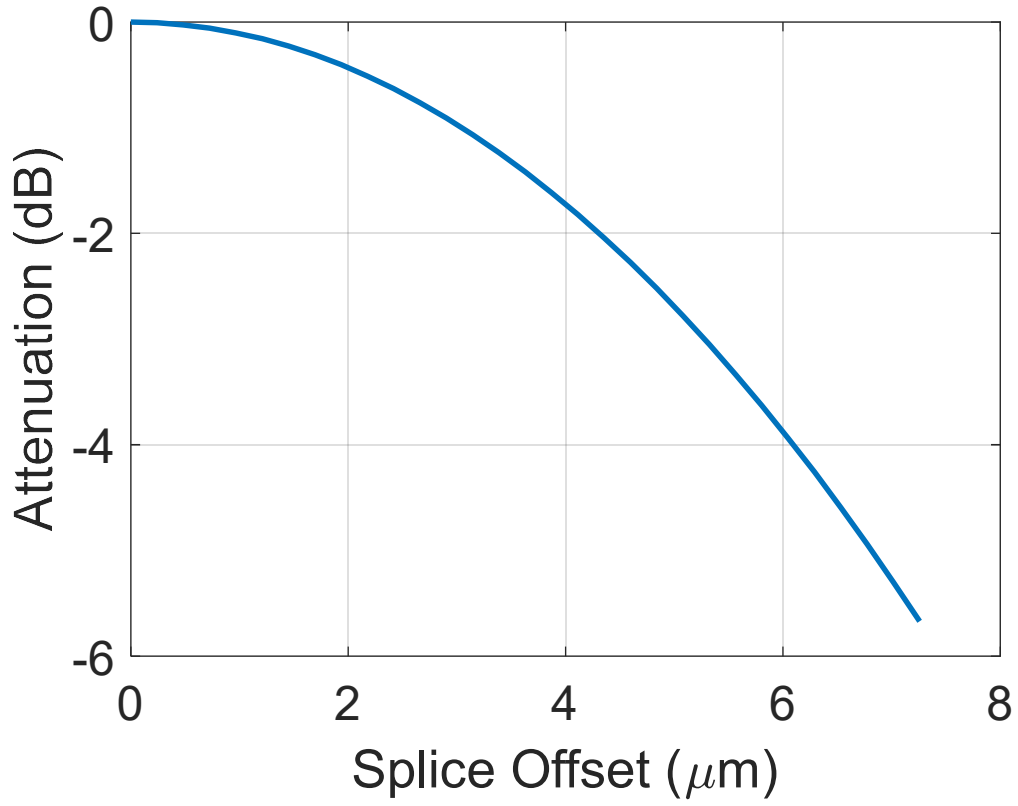
$$\psi_0 = \exp\left(-\frac{r^2}{w_2^2}\right)$$

$$\eta = \frac{\left|\int_0^\infty \exp\left(-r^2\left(\frac{1}{w_1^2}+\frac{1}{w_2^2}\right)\right) r\,dr\right|^2}{\int_0^\infty \exp\left(-\frac{2r^2}{w_1^2}\right) r\,dr \cdot \int_0^\infty \exp\left(-\frac{2r^2}{w_2^2}\right) r\,dr}$$

$$\int_0^\infty \exp(-ar^2)\,r\,dr = \frac{1}{2a} \quad \Rightarrow \quad \eta = \left(2\cdot\frac{w_1\cdot w_2}{w_1^2+w_2^2}\right)^2$$

$\rightarrow$ lateral offset Δx $\rightarrow$ $\eta = \left(2\cdot\frac{w_1 w_2}{w_1^2+w_2^2}\right)^2 \exp\left(-2\frac{\Delta x}{w_1^2+w_2^2}\right)$

Butt-coupling of (SI) SMF





University of Stuttgart
Germany

Concept Test

Concept test

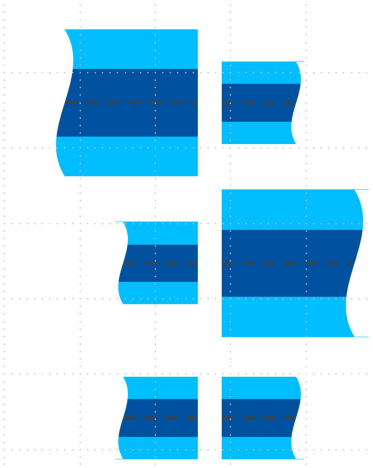
University of Stuttgart
Germany

Which fiber combination exhibit the smallest coupling loss ?
A = large core
B = small core

A A to B

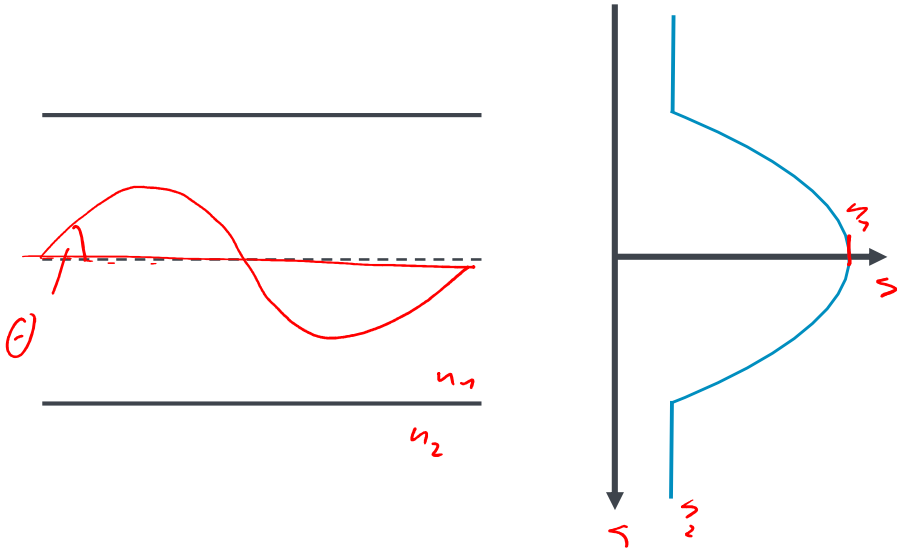
B B to A

C B to B



https://ilias3.uni-stuttgart.de/vote/EB5J

Profiledispersion



n : responsible for distance of ray

IV : responsible for speed of signal.

Profiledispersion

University of Stuttgart
Germany

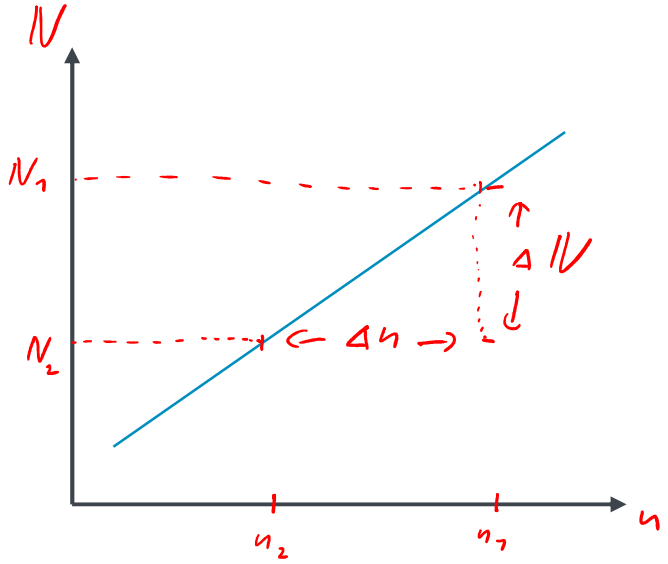
$$N(r) = \frac{N_1}{n_1}\left[(1-P)\cdot n(r) + n_1 \cdot P\right]$$

P: Profile-dispersion-parameter

$$\Delta n = \frac{n_1 - n_2}{n_1} \quad n_1 = \Delta \cdot n_1 \quad , \Delta = \frac{n_1 - n_2}{n_1}$$

$$N = n - \lambda \frac{dn}{d\lambda} .$$

$$\Rightarrow \Delta N = \Delta n - \lambda \frac{d(\Delta n)}{d\lambda}$$



Profiledispersion

$$\Delta N = \Delta \cdot n_1 - \lambda \frac{d(\Delta \cdot n_1)}{d\lambda}$$

$$= \Delta \cdot n_1 - \lambda \cdot \Delta \frac{d(n_1)}{d\lambda} - \lambda \cdot n_1 \cdot \frac{d\Delta}{d\lambda}$$

$$= \Delta N_1 - \lambda n_1 \frac{d\Delta}{d\lambda}$$

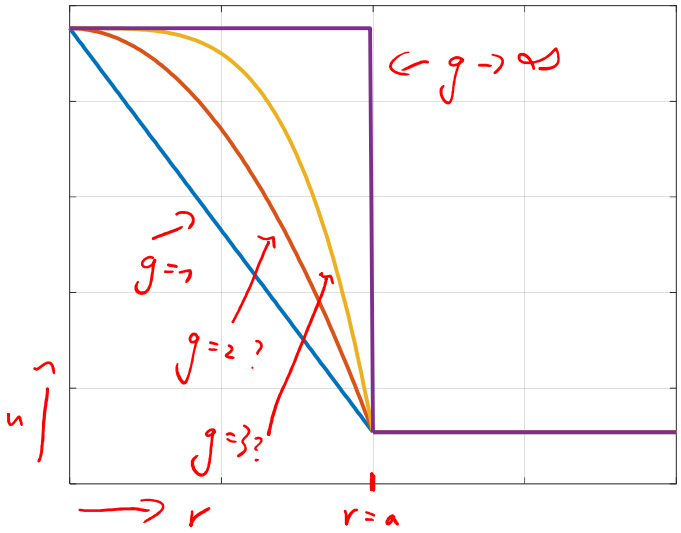
$$\Delta N = \frac{N_1}{n_1}(1-P)\Delta n = \Delta \cdot N_1 \cdot (1-P)$$

$$\rightarrow P = \frac{n_1}{N_1} \cdot \frac{\lambda}{\Delta} \cdot \frac{d\Delta}{d\lambda}$$

Profiledispersion

$$n^2(r) = \begin{cases} n_1^2\left(1 - 2\Delta\left(\frac{r}{a}\right)^g\right) & , r \le a \\ n_1^2(1-2\Delta) = n_2^2 & , r > a \end{cases}$$

$$\tau = \frac{d\beta}{d\omega} = \frac{1}{c} \cdot \frac{k_0}{\beta \cdot \frac{v_0^2}{2}} \int_0^{v_0} n(r)\, N(r)\, v\, dv$$



$$n(r)\,N(r) = \frac{N_1}{n_1}\left(\left(1-P\right)n^2(r) + n(r)\cdot n_1 \cdot P\right) \approx n_1 N_1\left(1 - 2\Delta\left(1-\frac{P}{2}\right)\left(\frac{r}{a}\right)^g\right)$$

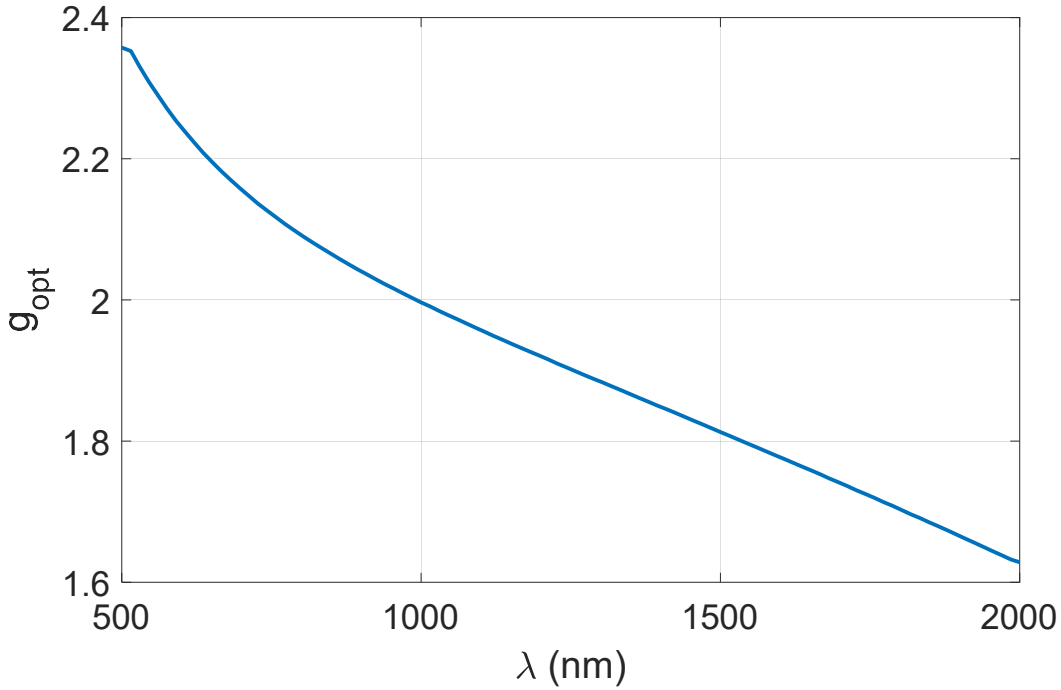
Profiledispersion

University of Stuttgart
Germany

$$\tau \approx \frac{1}{c} n_1 N_1 \left(1 - 4\Delta \frac{1-P_2}{g+2} \left(\frac{r_0}{a}\right)^g\right)$$

$$n_1 \left(1 - \Delta \left(\frac{r_0}{a}\right)^g\right)$$

$$g_{opt} = 2 - 2P$$

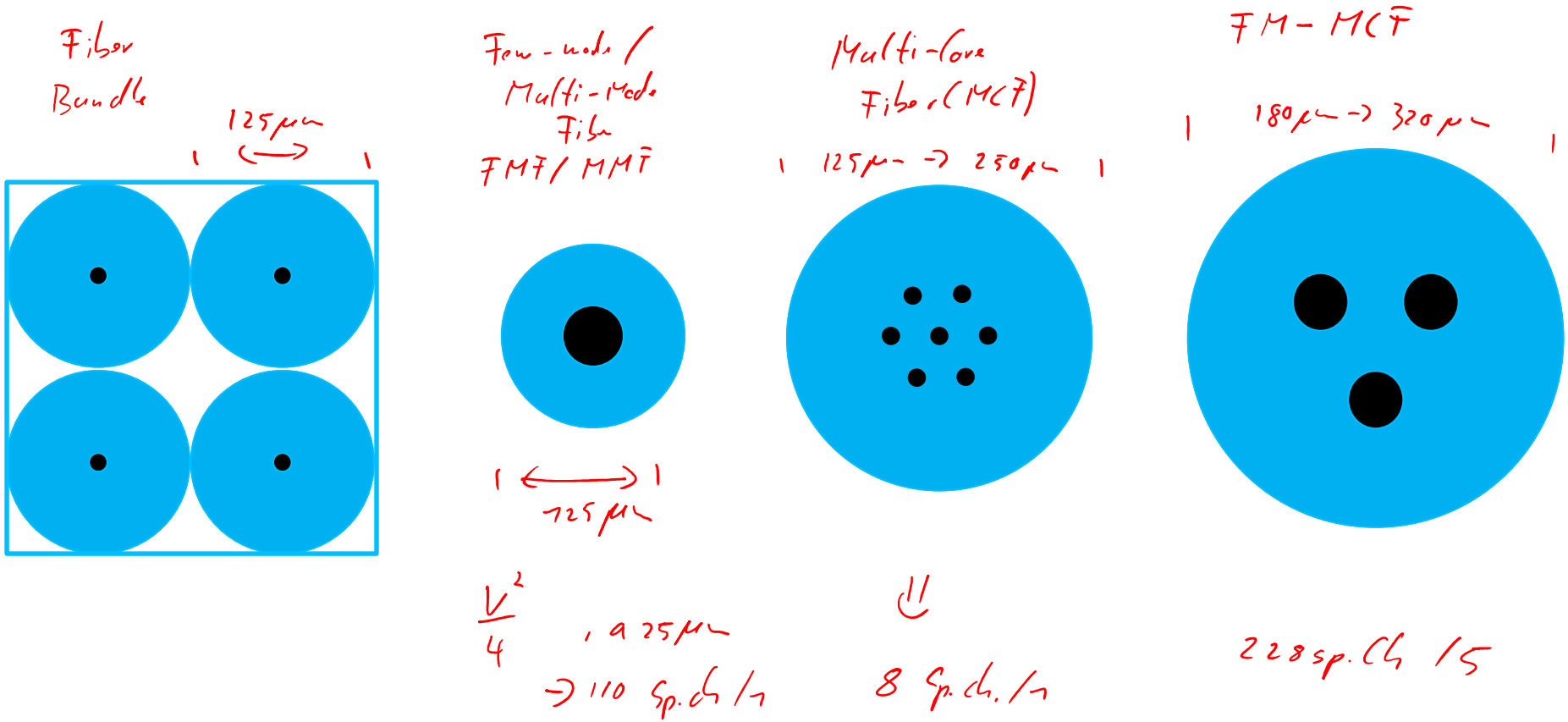


Different dispersion mechanisms

University of Stuttgart
Germany

Material dispersion	Waveguide dispersion	Profile dispersion
→ diff n → diff τ_{gr} → diff v_{gr}	→ can be used to compensate D_m	→ diff n → diff Δ
→ higher symbol rat → trouble	→ diff modes → diff v_{gr}	diff modes → diff v_{gr} → diff n
→ minimum ≈ 1300 nm	→ minimized using GI-Profile	→ can be minimized with correct g_{opt}

Optical fibers for space-division multiplexing





University of Stuttgart
Germany

Concept Test

Concept test

University of Stuttgart
Germany

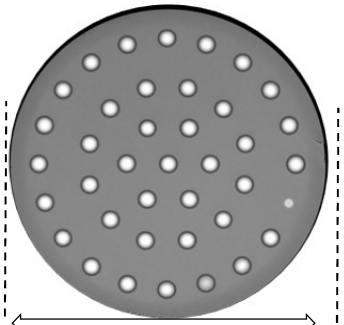
Which fiber type has the highest spatial information density ?

- **A** Multicore fiber
- **B** Few-mode-fiber
- **C** Multicore-few-mode- fiber

https://ilias3.uni-stuttgart.de/vote/IXN4

Record SDM fiber demonstrations

13 km 38-few-mode multi-core fiber (MCF)

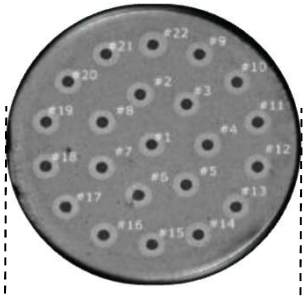


305 µm

2020
10.66 Pb/s
(C+L band)

2023
22 Pb/s
(S/C/L band)

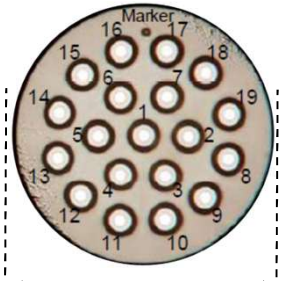
31.4 km 22-core MCF



260 µm

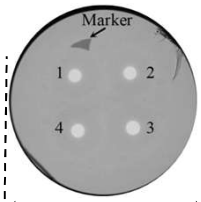
2015
2.15 Pb/s

30 km 19-core MCF



220 µm

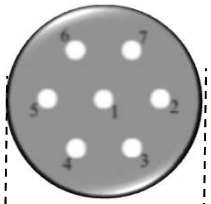
3.5 km 4-core few-mode MCF.



160 µm

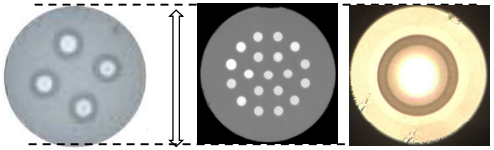
2018
1.2 Pb/s

53.7 km 7-core MCF



160 µm

54-100 km 4-core MCF



125 µm

2022
1.02 Pb/s

66 km randomly coupled MCF

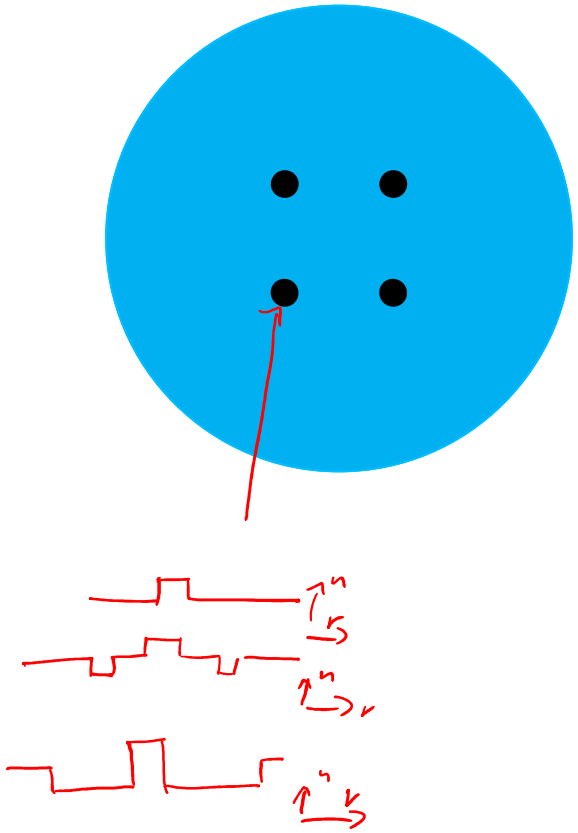
2023
1.7 Pb/s

26 km 55-mode fiber

2023
3.56 Pb/s

Weakly coupled Homogenous-core MCF

University of Stuttgart
Germany



Design and fabrication of ultra-low crosstalk and low-loss multi-core fiber

Tetsuya Hayashi,* Toshiki Taru, Osamu Shimakawa, Takashi Sasaki, and Eisuke Sasaoka

Optical Communications R&D Laboratories, Sumitomo Electric Industries, Ltd., 1 Taya-cho, Sakae-ku, Yokohama, 244-8588 Japan
**t-hayashi@sei.co.jp*

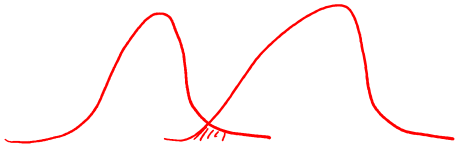
Multi-core fiber design and analysis: coupled-mode theory and coupled-power theory

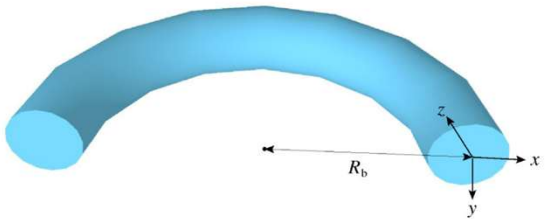
Masanori Koshiba,[1,*] Kunimasa Saitoh,[1] Katsuhiro Takenaga[2], and Shoichiro Matsuo[2]

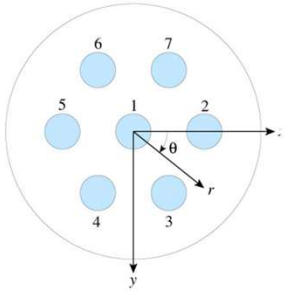
[1]Division of Media and Network Technologies, Hokkaido University, Sapporo 060-0814, Japan
[2]Optics and Electronics Laboratory, Fujikura Ltd., Sakura 285-8550, Japan
**koshiba@ist.hokudai.ac.jp*

Weakly coupled Homogenous-core MCF

University of Stuttgart
Germany







$$\frac{dA_m}{dz} = -j \sum_{n \neq m} \kappa_{mn} A_n(z) \cdot \exp(j \cdot \Delta\beta_{mn}) \cdot f(z)$$

$$\kappa_{mn} = \frac{1}{4} \sqrt{\frac{\varepsilon_0}{\mu_0}} \iint_A (n^2 - n_m^2) \psi_m^* \psi_n \, dA$$

$$f(z) = \exp(j(\phi_m - \phi_n)) \int f(z)$$

Weakly coupled Homogenous-core MCF

University of Stuttgart
Germany

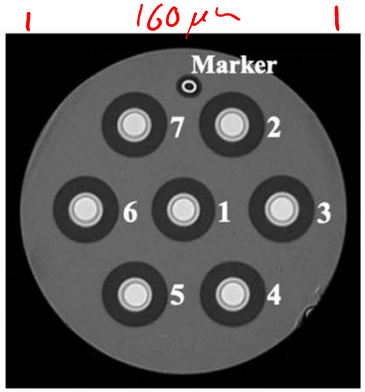


Fig. 9. Cross section of the fabricated MCF [16].

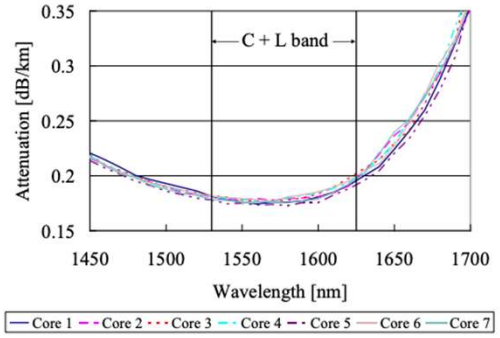


Fig. 10. Attenuation spectra of cores of the fabricated MCF.

Table 1. Designed Optical Properties of Each MCF Core at λ = 1550 nm

	Designed value
λ_{cc} [nm]	1496
MFD [μm]	9.86
A_{eff} [μm^2]	79.6
CD [ps/nm/km]	23.3
D. Slope [ps/nm^2/km]	0.063

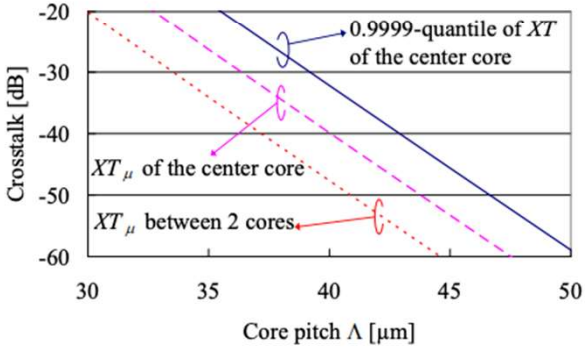
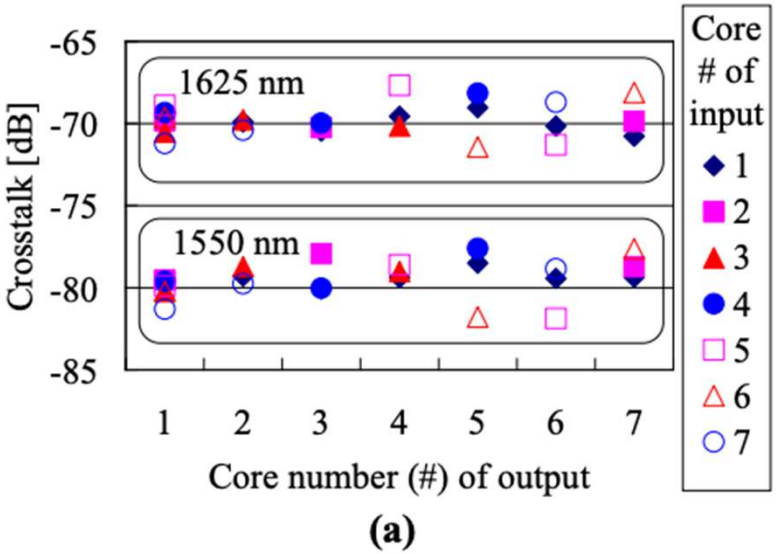


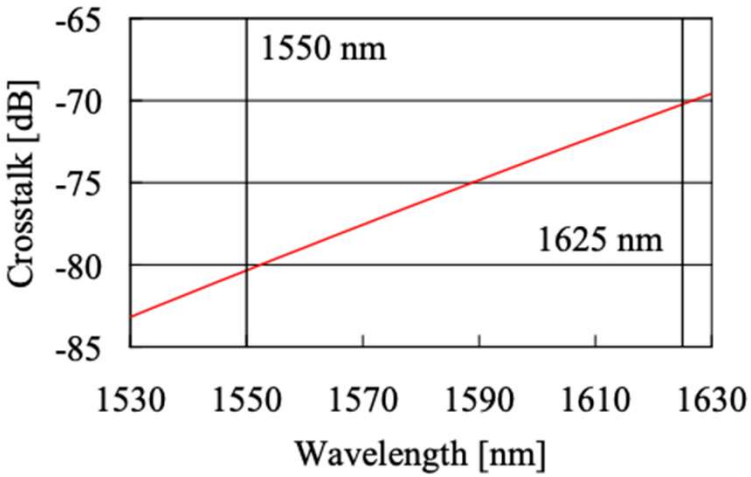
Fig. 7. Relationship between the core pitch Λ and the crosstalk for the designed core.

Weakly coupled Homogenous-core MCF

University of Stuttgart
Germany



(a)



(b)

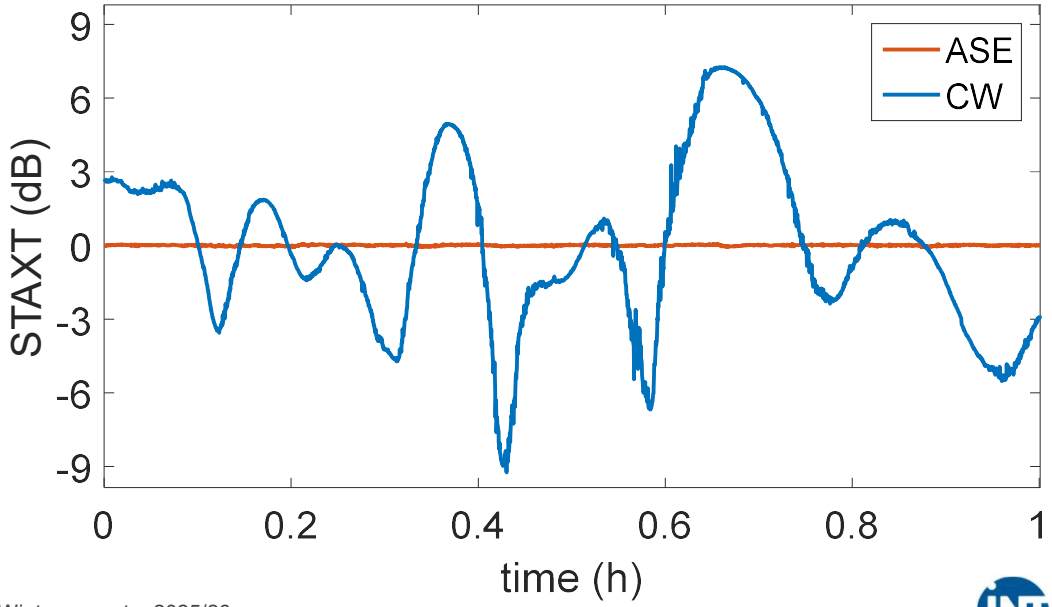
Research Article | Vol. 25, No. 10 | 15 May 2017 | OPTICS EXPRESS 12020
Optics EXPRESS

University of Stuttgart
Germany

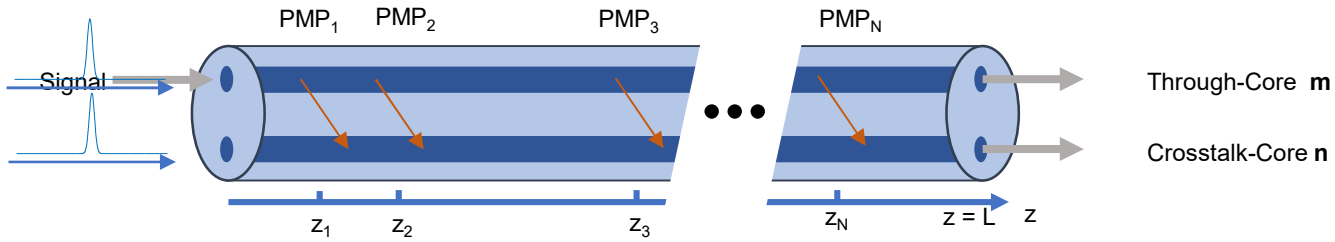
Crosstalk dynamics in multi-core fibers

GEORG RADEMACHER,* RUBEN S. LUÍS, BENJAMIN J. PUTTNAM, YOSHINARI AWAJI, AND NAOYA WADA

Photonics Systems Research Laboratory, National Institute of Information and Communications Technology (NICT), 4-2-1, Nukui-Kitamachi, Koganei, Tokyo, 184-8795, Japan
**georg.rademacher@nict.go.jp*



Modeling of inter-core crosstalk



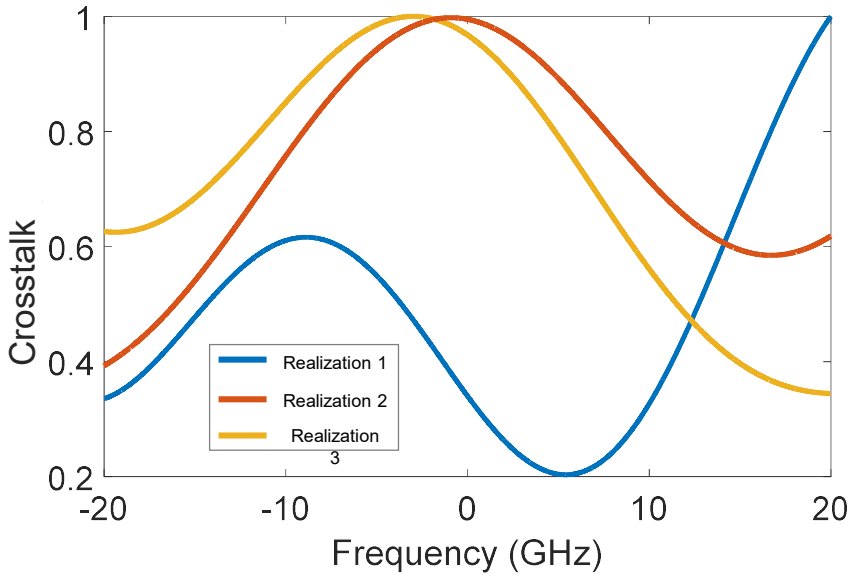
T. Hayashi et al., “Behavior of Inter-Core Crosstalk as a Noise and Its Effect on Q -Factor in Multi-Core Fiber,” IEICE Trans. Commun., 2014

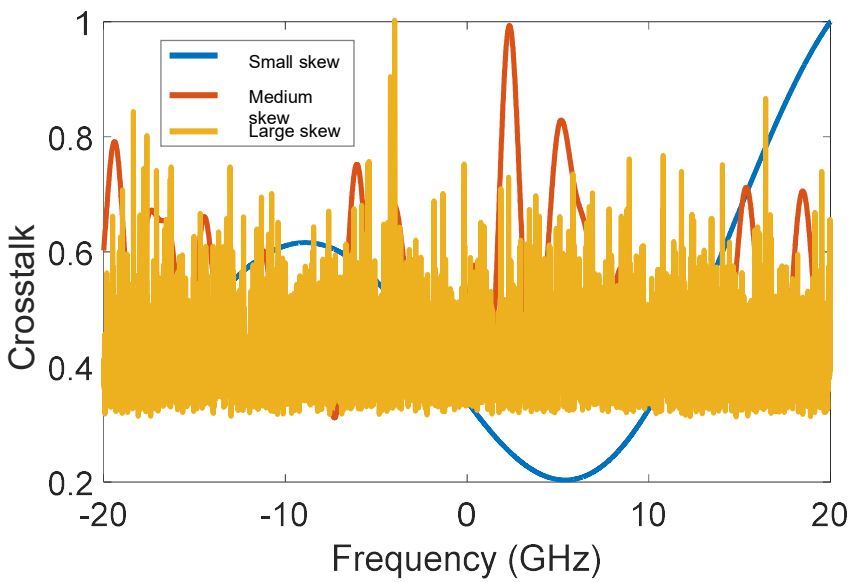
R. Luís et al., “Time and Modulation Frequency Dependence of Crosstalk in Homogeneous Multi-Core Fibers,” *JLT,* 2016

Influence of random phase / inter-core skew on transfer function

University of Stuttgart
Germany

$$H_{XT}(L,\omega) = -jK_{nm}\sum_{k=1}^{N} e^{-j\boldsymbol{\phi_k}} e^{-j s_{nm}\boldsymbol{z_k}\omega}$$

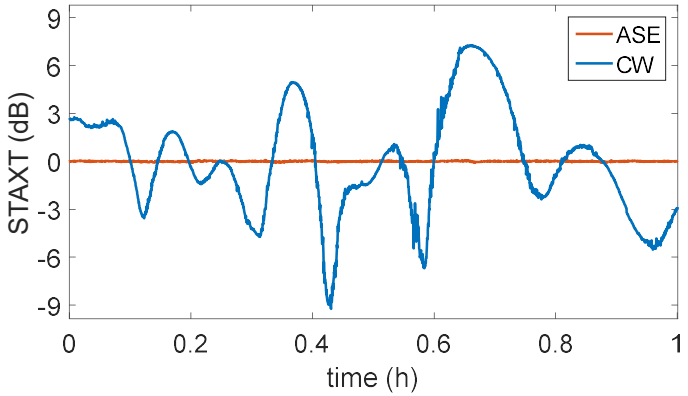




Variance of average crosstalk-power

University of Stuttgart
Germany

$$H_{XT}(L,\omega) = -jK_{nm}\sum_{k=1}^{N} e^{-j\boldsymbol{\phi_k}} e^{-js_{nm}\boldsymbol{z_k}\omega}$$

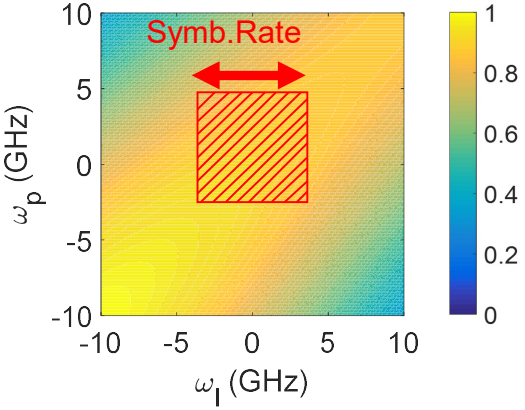


Variance of average crosstalk-power

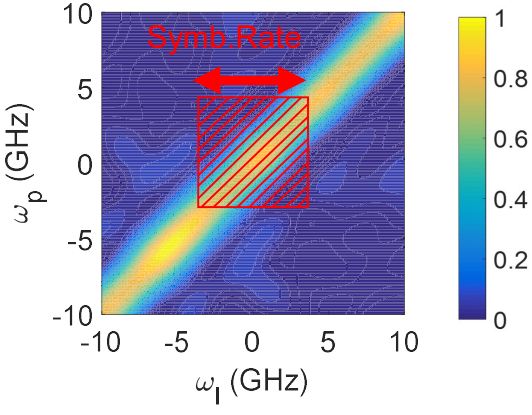
University of Stuttgart
Germany

$$\mathrm{VAR}_{\mathrm{XT}} \propto \sum_{l=1}^{M} \mathrm{VAR}(\boldsymbol{\Omega_l}) + \sum_{l \neq p}^{M} \sum_{p \neq l}^{M} \mathrm{COV}(\boldsymbol{\Omega_l}, \boldsymbol{\Omega_p})$$

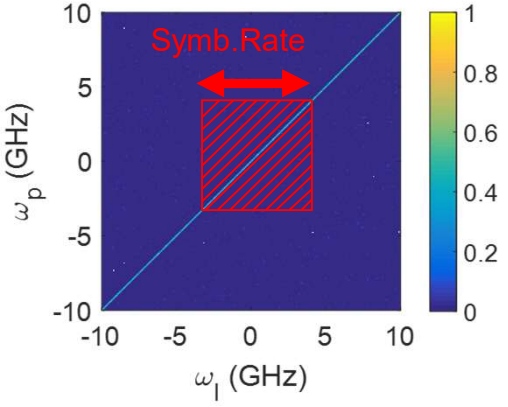
0.025 ns skew



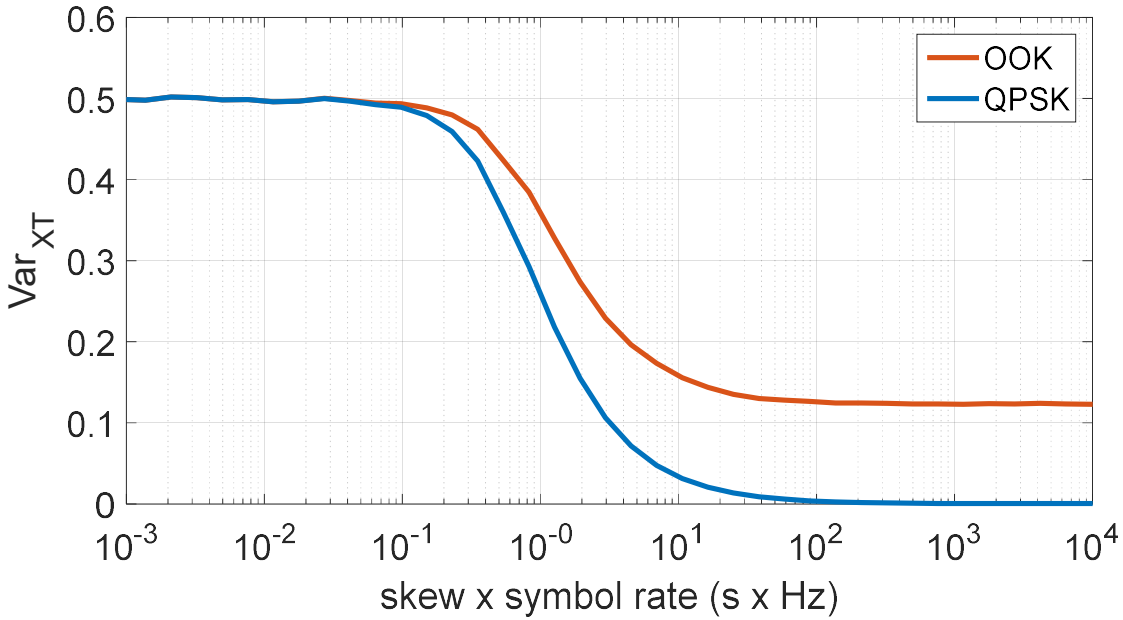
0.25 ns skew



250 ns skew



Variance of average crosstalk power





University of Stuttgart
Germany

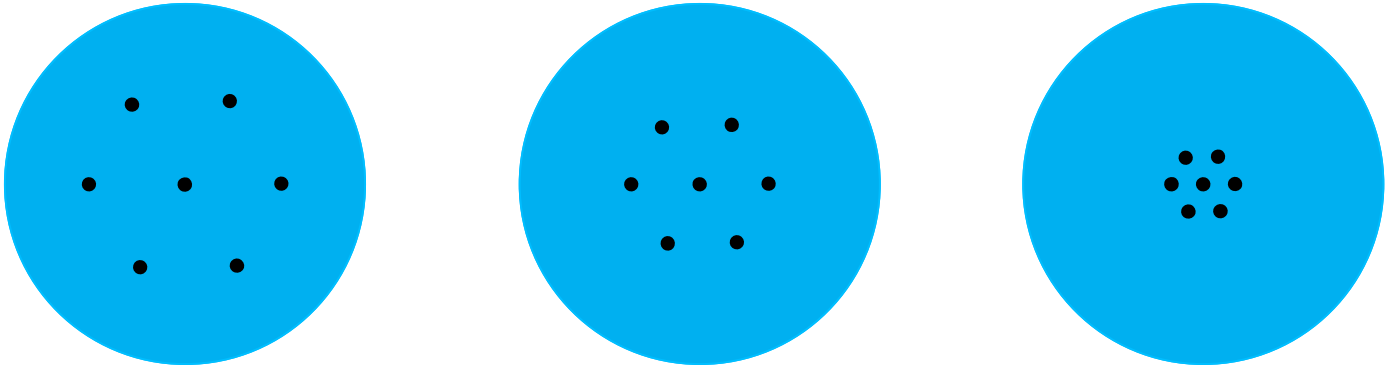
Concept Test

Randomly-Coupled Multi-Core Fiber Technology

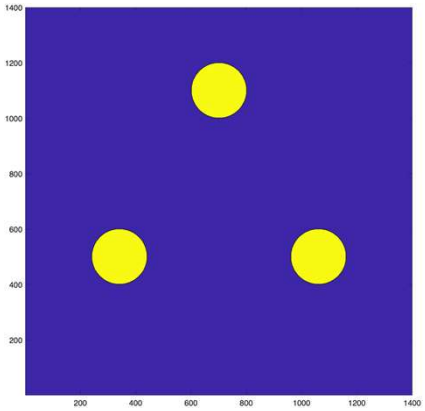
This article considers an even higher density spatially parallel integration technique that allows for random coupling between the cores of multi-core fibers.

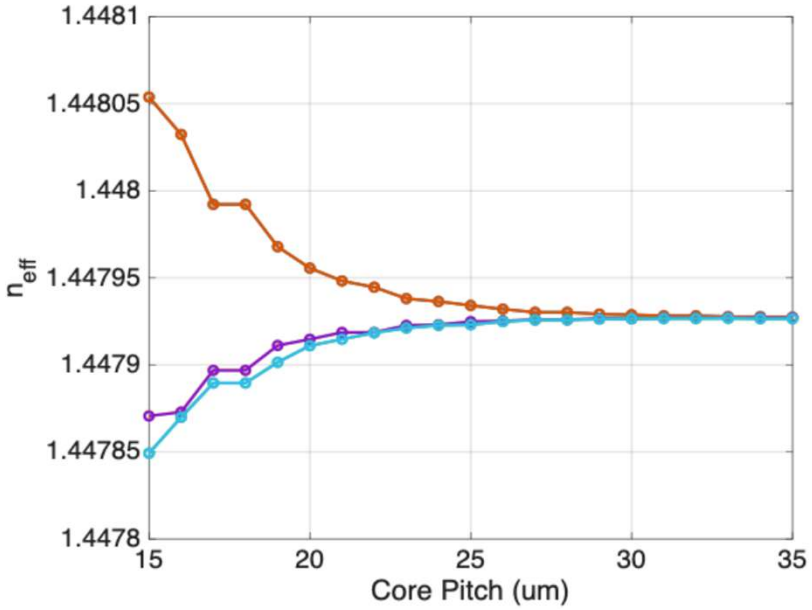
By Tetsuya Hayashi, *Senior Member IEEE*, Taiji Sakamoto, *Member IEEE*, Yusuke Yamada, Roland Ryf, *Fellow IEEE*, René-Jean Essiambre, *Fellow IEEE*, Nicolas Fontaine, *Senior Member IEEE*, Mikael Mazur, *Member IEEE*, Haoshuo Chen, *Senior Member IEEE*, and Takemi Hasegawa, *Member IEEE*

Properties for CC-MCF

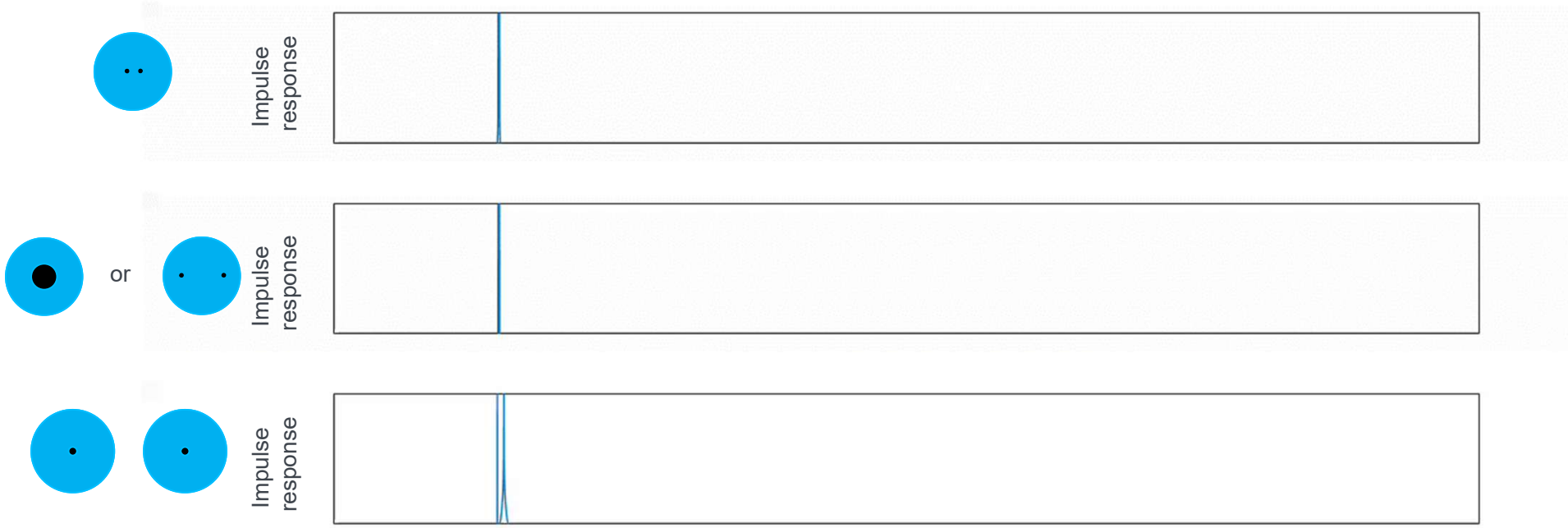


Properties for CC-MCF





Temporal spread of signals in SDM fibers

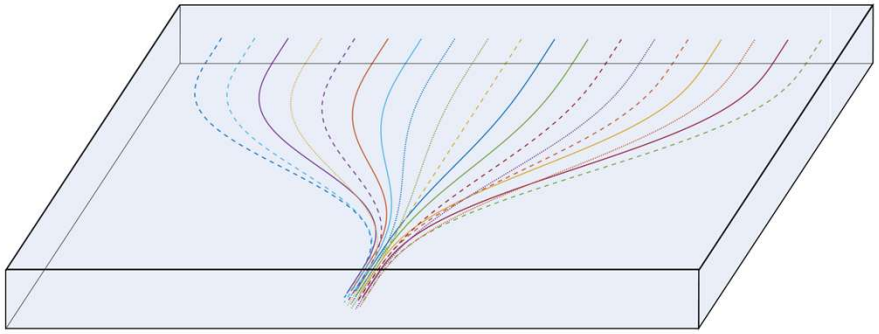


Randomly Coupled 19-Core Multi-Core Fiber with Standard Cladding Diameter

Georg Rademacher[(1)], Menno van den Hout[(1,2)], Ruben S. Luís[(1)], Benjamin J. Puttnam[(1)], Giammarco Di Sciullo[(1,3)], Tetsuya Hayashi[(4)], Ayumi Inoue[(4)], Takuji Nagashima[(4)] Simon Gross[(5)], Andrew Ross-Adams[(6)], Michael J. Withford[(6)], Jun Sakaguchi[(1)], Cristian Antonelli[(3)], Chigo Okonkwo[(2)], Hideaki Furukawa[(1)]

[(1)] NICT, 4-2-1, Nukui-Kitamachi, Koganei, Tokyo, 184-8795, Japan,
[(2)] High Capacity Optical Transmission Lab, Eindhoven University of Technology, Eindhoven, The Netherlands
[(3)] University of L'Aquila and CNIT, 67100, L'Aquila, Italy
[(4)] Sumitomo Electric Industries, Ltd., 1 Taya-cho, Sakae-ku, Yokohama 244-8588, Japan
[(5)] MQ Photonics Research Centre, School of Engineering, Macquarie Univ., Sydney, Australia.
[(6)] MQ Photonics Research Centre, School of Math. and Physical Sciences, Macquarie Univ., Sydney, Australia.
georg.rademacher@nict.go.jp

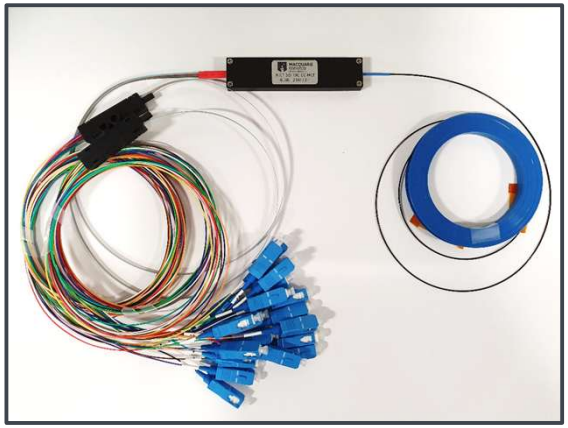
Input single mode fiber array



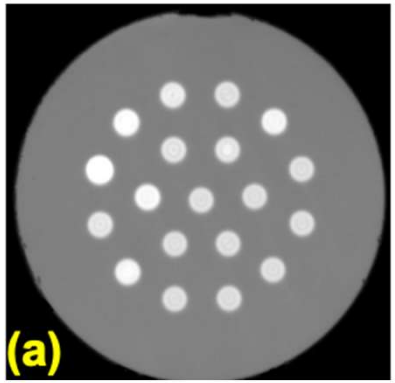
Output RC-19-MCF

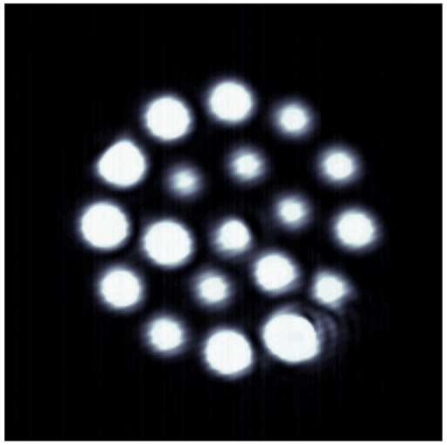
S. Gross and M.J. Withford, "Ultra fast-laser-inscribed 3D integrated photonics," Nanophotonics 4 (1) (2015)

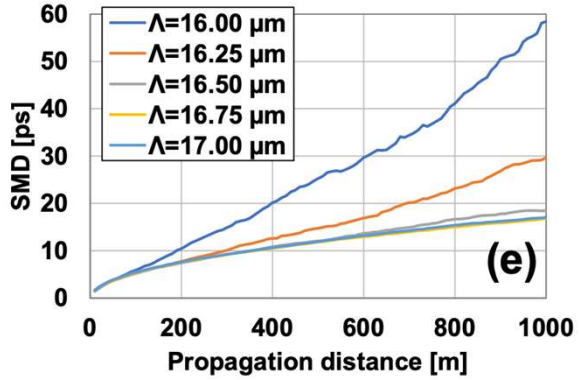


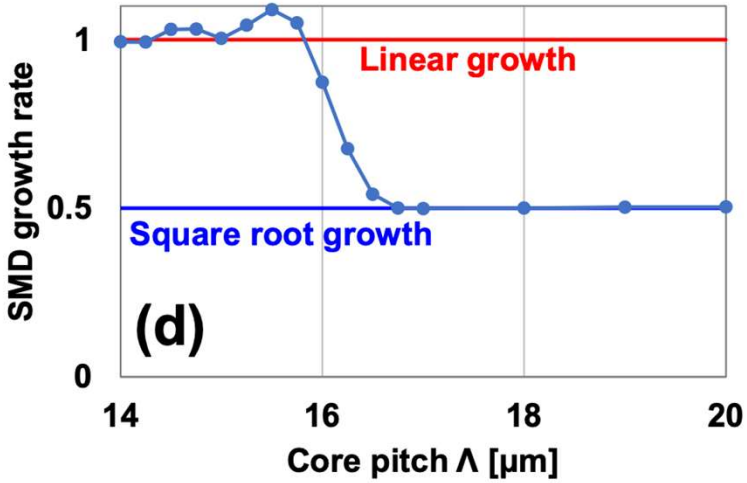


Geo
rg


(a)




Λ=16.00 µm
Λ=16.25 µm
Λ=16.50 µm
Λ=16.75 µm
Λ=17.00 µm
SMD [ps]
0
10
20
30
40
50
60
0
200
400
600
800
1000
Propagation distance [m]
(e)


SMD growth rate
0
0.5
1
Linear growth
Square root growth
(d)
14
16
18
20
Core pitch Λ [µm]

University of Stuttgart
Institute of electrical and optical communications

Thank you!



Prof. Dr.-Ing. Georg Rademacher

e-mail georg.rademacher@int.uni-stuttgart.de
phone +49 (0) 711 6856 7901
www. int.uni-stuttgart.de/

University of Stuttgart
Institute of electrical and optical communications

